

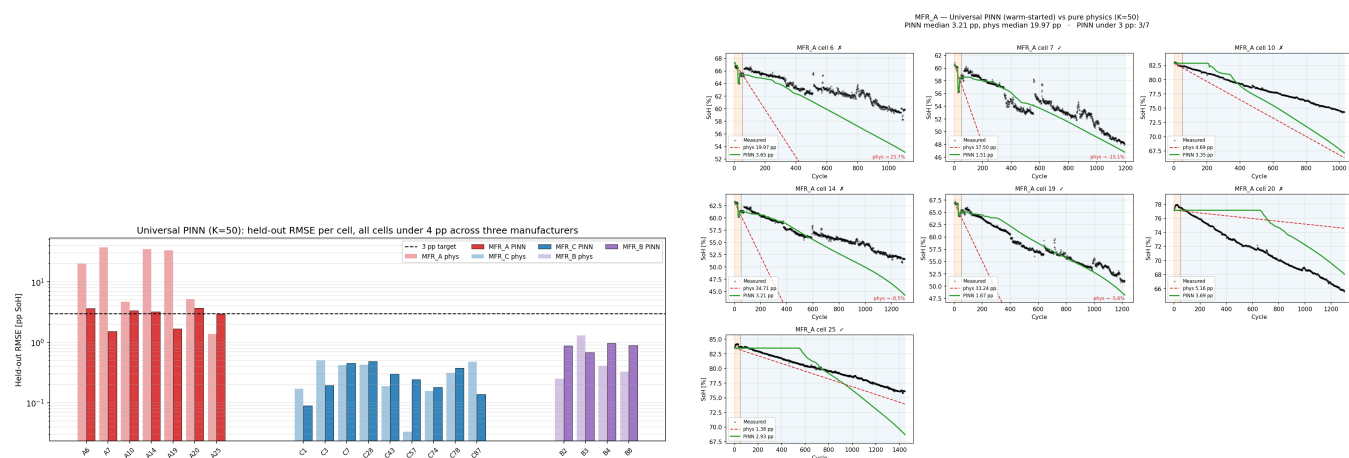
50-cycle characterisation across three manufacturers, 8× cycling reduction vs pure PyBaMM

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Motivation. Second-life repurposing of used LFP cells for stationary energy storage is bottlenecked by characterisation cost: pack builders must estimate remaining useful life (RUL) with no beginning-of-life data and limited cycling. Pure PyBaMM [1] reaction-limited SEI calibration requires ≈ 400 cycles per cell for < 3 pp SoH extrapolation error on used cells. Prior second-life optimisation [3] and universal-differential-equation degradation [4] work improves calibration quality without reducing the cycling budget. We ask: *can one physics-informed neural network with fixed hyperparameters cut required characterisation across LFP manufacturers?*

Method. A joint PINN represents SoH as $\text{SoH}_\theta(n) = \text{SoH}_{\text{init}} - \text{softplus}(\text{NN}_\theta)$ (hard-monotonic decrement) with a per-cell latent embedding. Physics constraint: $d\text{SoH}/dn = -k_{\text{SEI}}(i) \text{SoH}^{p(i)}$ (Ramadass-canonical rxn-limited SEI [5]), with per-cell learnable (k_{SEI}, p) warm-started from a linear fit on the training window. A brief NN warm-start phase pre-shapes the network to the initial linear-fade slope before the physics loss engages, eliminating a flat-start artefact that otherwise degrades deep-fade cells at small K . Loss combines data MSE, physics MSE on collocation points across the full cycle domain, boundary anchoring, and monotonicity regularisation. Adam + cosine schedule, $\lambda_{\text{phys}}=2$, ≈ 9 min per run on a single GPU.

Result. 20 LFP cells across three manufacturers (7 MFR_A, 9 MFR_C, 4 MFR_B), *identical* architecture and hyperparameters ($K=50$, $\approx 68k$ params): the joint PINN achieves held-out RMSE < 4 pp on **all 20 cells** and < 3 pp on 16/20 cells (Fig. 1a). On deep-second-life MFR_A cells (15–25 pp fade over 1000+ cy) where pure physics fails catastrophically (median 20.0 pp), the PINN maintains **median 3.2 pp** (mean 2.9 pp, max 3.7 pp) — a **6× reduction** with 7/7 head-to-head wins (Fig. 1b). On near-fresh MFR_C/MFR_B cells (1–3 pp fade) physics is already near-optimal; the PINN matches or improves (MFR_C median 0.24 pp, MFR_B median 0.89 pp). **Vs the pure-physics $K=400$ baseline, the same PINN reaches equivalent engineering accuracy at $K=50$ — 8× cycling reduction with make-agnostic transfer.**



(a) Per-cell held-out RMSE across 3 manufacturers (log scale). MFR_A physics towers above 3 pp target; PINN sits at or below 3.7 pp on every cell.

(b) MFR_A trajectories ($K=50$). Physics dashed lines exit the frame with impossible-SoH annotations; PINN green tracks measured across 1000+ cycles.

Scope & PyBaMM usage. Results hold for LFP at 25 °C isothermal. Cells excluded by upstream shape filter (non-monotonic, three-regime, batch-transition data artefacts) remain out of scope — PINN cannot recover cells whose canonical trajectories don't correspond to a coherent lifetime. PyBaMM v26.4.3 provides the reference chemistry (Prada2013); its analytical rxn-limited SEI rate is embedded as the physics constraint.

References. [1] V. Sulzer et al., *PyBaMM*, J. Open Res. Softw. 9(1) 14–21 (2021). [2] M. Prada et al., *LiFePO₄-graphite Li-ion model*, J. Electrochem. Soc. 160(4) A616–A628 (2013). [3] O. Ouledboutaarija et al., *Advancing Second-Life Batteries*, PyBaMM Conf. 2025, p. 19. [4] J. A. Kuzhiyil et al., *Generalisability via UDE*, PyBaMM Conf. 2025, p. 12. [5] P. Ramadass et al., *Capacity fade first-principles model*, J. Electrochem. Soc. 151(2) A196–A203 (2004).